

Small Aircraft Maintenance Schedule Approval

Approbation du calendrier de maintenance pour les petits aéronefs

Operator – Exploitant Orizon Aviation Québec		Aircraft type/model(s): – Type/modèle(s) de l'aéronef : Cessna 172 N and P	
Type of operation – Type d'exploitation ☒ Flying training operations pursuant to CAR IV Formation au pilotage, conformément au RAC IV ☒ Commercial operations pursuant to CAR VII Opérations commerciales, conformément au RAC VII ☐ Private operation pursuant to CAR VI Exploitation privée, conformément au RAC VI			
Aircraft role(s) – Rôle(s) de l'aéronef Formation au pilotage, travail aérien et Taxi aérien			
Annual utilization ¹ Utilisation annuelle ¹	Min		Max
	Hr. – Heures 400	Cycles	Hr. – Heures 700
This approval is conditional upon the information specified above. In the event an aircraft's actual annual utilization is outside the range specified, or type of operation or aircraft role differs from that stated, the operator must undertake a review of this schedule, identify any amendments necessary to cater for the change in circumstances, and obtain Transport Canada approval to incorporate those amendments.		Cette approbation dépend des renseignements énoncés ci-dessus. Si l'utilisation annuelle réelle de l'aéronef dépasse la limite précisée, ou si le type d'exploitation ou le rôle de l'aéronef diffère de celui qui est énoncé, l'exploitant doit entreprendre un examen de ce calendrier, identifier les modifications nécessaires pour tenir compte des changements de conditions et obtenir de Transports Canada l'autorisation d'incorporer ces modifications.	
Signature of Operator Signature de l'exploitant		2020-12-08 Date (yyyy - mm - dd) Date (aaaa - mm - jj)	
		Q-0800	
For Minister of Transport Pour le ministre des Transports		Date (yyyy - mm - dd) Date (aaaa - mm - jj)	
		Transport Canada Approval No. N° d'approbation de Transports Canada	

Revision status (Transport Canada use only) ² – État de la révision (À l'usage de Transports Canada seulement) ²									
Pages	<u>4</u>	Pages	<u>4,5,6</u>	Pages	<u>4,5,</u>	Pages	<u>5</u>	Pages	<u>1 à 20</u>
Rev. Rév.	<u>1</u>	Rev. Rév.	<u>2</u>	Rev. Rév.	<u>3</u>	Rev. Rév.	<u>4</u>	Rev. Rév.	<u>5</u>
Date ³	<u>2008-05-22</u>	Date ³	<u>2008-12-09</u>	Date ³	<u>2011-09-01</u>	Date ³	<u>2011-10-26</u>	Date ³	<u>2016-04-27</u>
Pages	<u>1 à 20</u>	Pages	<u>1,3,5</u>	Pages	<u>1 à 17 (all)</u>	Pages		Pages	
Rev. Rév.	<u>6</u>	Rev. Rév.	<u>7</u>	Rev. Rév.	<u>8</u>	Rev. Rév.		Rev. Rév.	
Date ³	<u>2019-10-30</u>	Date ³	<u>2020-09-30</u>	Date ³	<u>2020-12-08</u>	Date ³		Date ³	

¹ Complete this section only where the maintenance schedule approval is predicated upon an anticipated level of utilization.

Ne remplir cette section que si l'approbation du calendrier de maintenance est fondée sur le niveau prévu d'utilisation.

² Revision section refers to all pages in the approved schedule, including this approval document. Where the same page is referenced in more than one block, the most recent revision indicated supersedes all earlier references.

La section révision renvoie à toutes les pages du calendrier approuvé, y compris le présent document d'approbation. Lorsque l'on renvoie à la même page plus d'une fois dans le même bloc, la révision la plus récente indiquée remplace toutes les autres références.

³ Please use date format of (yyyy - mm - dd).

S'il vous plaît utilisez le format du date (aaaa - mm - jj).

General Conditions

- This document, together with the additional pages referenced herein, constitutes the minimum scheduled maintenance to be performed. Nothing contained in, or omitted from the maintenance schedule absolves the operator from the responsibility for ensuring the aircraft are maintained in an airworthy condition.
- Nothing in this document shall be construed as exempting the operator from responsibility for compliance with all applicable component life limits, Airworthiness Limitations, or other mandatory requirements.
- The operator shall ensure that all tasks listed in the currently approved revision of this schedule are completed within the intervals specified.
- The operator shall amend the schedule as and when directed by the Minister, and shall evaluate all recommendations issued in the form of Instructions for Continued Airworthiness (ICAs) and/or publications issued by the Type Certificate (TC) holder (airframe, engine, or propeller). These instructions need to be evaluated for applicability to an air operator's program in accordance with standard 726.07 of the CARs titled Air Operator Maintenance - Evaluation Program. Where appropriate, the operator shall initiate amendment action.
- Amendments to this schedule must be approved for any change in type of operation or operating role identified overleaf and, where maximum and minimum utilization data are specified, for any variation outside the limits indicated. Approval is also required for any task deletions, increases in intervals, or other significant changes. Transport Canada approval is not required for amendments that involve only the addition of tasks or a reduction of intervals; however the operator shall notify Transport Canada of these changes within 30 days.
- The operator shall ensure that maintenance personnel are provided with such records and other documents as are necessary to enable them to determine to their satisfaction, that the aircraft is in compliance with the airworthiness requirements applicable to the work undertaken. All work required by this schedule shall be released in accordance with CAR 571.10.
- The tolerances specified in this schedule shall not apply to any Airworthiness Limitations or Airworthiness Directives. Exceptions or deviations from this maintenance schedule must be submitted into Transport Canada for approval, together with substantiating data.

Conditions générales

- Ce document, ainsi que les pages supplémentaires indiquées ici, constituent la maintenance planifiée minimale à exécuter. Rien de ce qui est contenu ou omis dans le calendrier de maintenance ne décharge l'exploitant de la responsabilité de s'assurer que l'aéronef est maintenu en état de navigabilité.
- Rien dans ce document ne doit être interprété comme une exonération de la responsabilité de l'exploitant à l'égard de la conformité à toutes les limites de la durée de vie des composants, des limitations de la navigabilité et autres exigences obligatoires.
- L'exploitant doit faire en sorte que toutes les tâches énumérées dans la révision approuvée actuelle de ce calendrier sont exécutées dans les délais fixés.
- L'exploitant doit modifier le calendrier selon les directives du ministre et évaluer toutes les recommandations contenues dans les instructions de maintien de la navigabilité ou dans les publications délivrées par le titulaire du certificat de type (cellule, moteur ou hélice). Ces instructions doivent être évaluées en fonction du programme d'un exploitant aérien conformément à la norme 726.07 du RAC - Programme d'évaluation. Le cas échéant, l'exploitant prendra des dispositions pour effectuer les modifications.
- Les modifications apportées à ce calendrier doivent être approuvées pour tout changement de type d'exploitation ou du rôle de l'aéronef indiqués au verso, lorsque les données sur l'utilisation maximale et minimales sont précisées, pour tout écart par rapport aux limites indiquées. Il faut également une approbation pour toute suppression de tâche, augmentation des intervalles ou autres changements importants. L'approbation de Transports Canada n'est pas nécessaire pour les modifications qui impliquent l'ajout de tâches ou une réduction des intervalles uniquement; toutefois, l'exploitant doit avertir Transports Canada de ces changements dans un délai de 30 jours.
- L'exploitant doit faire en sorte que le personnel de la maintenance reçoive tous les dossiers et autres documents nécessaires pour lui permettre de déterminer, à sa satisfaction, que l'aéronef est conforme aux exigences de navigabilité qui s'appliquent aux travaux effectués. Tous les travaux exigés par ce calendrier feront l'objet d'une certification après maintenance, conformément au RAC 571.10.
- Les tolérances précisées dans ce calendrier ne s'appliquent pas aux limitations de navigabilité, ni aux circulaires de navigabilité.
- Les exceptions et les écarts par rapport à ce calendrier de maintenance, ainsi que la documentation justificative, doivent être présentés à Transports Canada afin d'obtenir une approbation.

Scheduled Check Cycle
Cycle de vérification prévu

The aircraft will be inspected in accordance with the check cycle specified in table 1 below. Check intervals may be varied within the tolerances specified. Detailed instructions and procedures for scheduled maintenance are contained in the attached check list (the pages of which are identified in the revision status block) or in maintenance schedule reference....Q-0800..... rev #...8.....

L'aéronef doit être inspecté conformément au cycle de vérification précisé au tableau 1 ci-dessous. Les intervalles de vérification peuvent varier à l'intérieur des tolérances fixées. Les instructions et les procédures détaillées relatives au calendrier de maintenance figurent dans la liste de vérification ci-jointe (les pages en sont identifiées dans le bloc sur l'état de la révision) ou dans le renvoi au calendrier de maintenanceQ-0800.....n° de rév8.....

Table 1 - Check Cycle
Tableau 1 - Cycle de vérification

[illegible]

Notes

(Use this section if necessary, to explain the operation of the check cycle)
Utiliser cette section au besoin pour expliquer le fonctionnement du cycle de vérification)

Note 1 : Les inspections sont effectuées dans une fenêtre de plus ou moins 10 hrs selon l'intervalle du tableau ci-haut.

Note 2 : Toutes les tâches hors calendrier (tableau 3) seront synchronisées selon leur calendrier avec la prochaine inspection prévue.

Out of Phase Tasks & Equipment Maintenance Requirements

Tâches hors calendrier et exigences de maintenance de l'équipement

Engine & propeller overhauls and other maintenance tasks scheduled to occur out of phase with the inspection check cycle, shall be performed as indicated in table 3 below. Where applicable, the tasks may be identified by reference to separate documents, provided the documents are listed in table 2. Any out of phase tasks not listed shall be performed at the intervals specified in STD 625, Appendix C.

Note: Reference to other documents or to STD 625, Appendix C, does not relieve the owner/operator from the responsibility for determining the applicability of the tasks and intervals concerned, nor from the responsibility for identifying any other applicable maintenance requirements not listed therein. La révision des moteurs et des hélices et autres tâches de maintenance devant être effectuées en dehors du cycle des inspections sont exécutées conformément aux indications figurant au tableau 3 ci-dessous. Le cas échéant, on peut identifier les tâches en se reportant à des documents séparés, à condition qu'ils soient indiqués au tableau 2. Toutes les tâches hors calendrier qui ne sont pas indiquées seront exécutées aux intervalles précisés dans l'appendice C de la norme 625.

Remarque : Le renvoi à d'autres documents ou à l'appendice C de la norme 625 ne décharge pas le propriétaire/exploitant de la responsabilité de déterminer l'applicabilité des tâches et des intervalles en cause, ni de la responsabilité d'identifier les autres exigences de maintenance applicables ne figurant pas ici.

Table 2 - Reference documents

Tableau 2 - Documents de référence

Document Name Titre de document	Reference Number N° de référence	Revision Number N° de révision

Table 3 - Out of phase tasks

Tableau 3 - Tâches hors calendrier

Out of phase tasks & equipment maintenance requirements ⁴ Tâches hors calendrier et exigences de maintenance de l'équipement ⁴			
Item Article	Task Tâche	Interval ⁵ Intervalle ⁵	Tolerance Tolérance
Engines* O-320-H2AD Moteurs* O-320-D2J / O-360-A4M	Overhaul Révision	2200 HRS 2400 HRS	100 HRS 100 HRS
Propellers* Hélices* HÉLICE À PAS FIXE	CORROSION INSPECTION	5 YEARS	3 MONTHS
ALTERNATOR	INSPECT BRUSHES LEAD AND SLIPRING (J12)	500 HRS	50 HRS AND/OR NOTE 2
ALTIMETER	TEST AND CALIBRATION AWM 571 APP B	24 MONTHS	2 MONTHS
BEACON LIGHT SWITCH	REPLACE	ON CONDITION	N/A
BRACKET AIR FILTER	INSPECT REPLACE	100 HRS 12 MONTHS	NOTE 2 1 MONTH
COMPASS	CALIBRATE	12 MONTHS	1 MONTH
ELT	TEST AND CERTIFIED AS PER CAR'S 571 APP. G	12 MONTHS	1 MONTH
ELT BATTERY	REPLACE	REC. DU MANUF.	1 MONTH
ENCODER	TEST AND CALIBRATION AWM 571 APP F	24 MONTHS	2 MONTHS

⁴ Include additional pages where required. Ajouter des pages supplémentaires, au besoin.

⁵ Insert interval, specifying whether in hours, cycles or calendar time

Insérer les intervalles en précisant s'il s'agit d'heures, de cycles ou de périodes déterminées.

* Completion of engine and propeller details is mandatory. If applicable, indicate "On-condition."

Il est obligatoire de fournir des détails sur les moteurs et les hélices. Le cas échéant, indiquer « en état de marche ».

Out of phase tasks & equipment maintenance requirements (continued)
Tâches hors calendrier et exigences de maintenance de l'équipement (suite)

Item Article	Task Tâche	Interval Intervalle	Tolerance Tolérance
ENGINE CONTROLS	REPLACE	4800 HRS	100 HRS AND/OR NOTE 2
ENGINE HOSES	REPLACE	10 YEARS	3 MONTHS
FIRE EXTINGUISHER	INSPECT TEST HYDROSTATIC	12 MONTHS 12 YEARS	1 MONTH 1 MONTH
FIRST AID KIT	INSPECT CONTENT IF SEAL BROKEN OR REPLACE TIME X ITEM	12 MONTHS	1 MONTH
FUEL TANK AND SCREEN	DRAIN, INTERIOR CHECK AND ATTACHEMENT	1000 HRS	100 HRS AND/OR NOTE 2
FUEL QUANTITY INDICATORS	INSPECT FOR DAMAGE (K8)	12 MONTHS	1 MONTH
GYRO CENTRAL AIR FILTER	REPLACE	500 HRS	100 HRS AND/OR NOTE 2
LANDING LIGHT SWITCH	REPLACE	ON CONDITION	N/A
MAGNETOS (INTERNAL)	INSPECT	500 HRS	50 HRS AND/OR NOTE 2
STARTER	INSPECT (J13)	500 HRS	50 HRS AND/OR NOTE 2
STC : SA01695SE DOC NUMBER 190-00673-01 REV. F	VISUAL INSPECTION ANTENNA AND MOUNTING/WHEN INSTALLED ON THE AIRCRAFT	12 MONTHS	1 MONTH
STC : SA0181WI	G5 BATTERY CAPACITY TEST/ WHEN INSTALLED ON THE AIRCRAFT	12 MONTHS	1 MONTH
STC : SE1779NM SVS 5	INSPECT ACCORDING STC/WHEN INSTALLED ON THE AIRCRAFT	12 MONTHS 24 MONTHS	1 MONTH 1 MONTH
TACHOMETER	ACCURACY CHECK	12 MONTHS	1 MONTH
TAXI LIGHT SWITCH	REPLACE	ON CONDITION	N/A
TRANSPONDER	TEST AND CALIBRATION AWM 571 APP F	24 MONTHS	2 MONTHS
TRIM TAB ACTUATOR	LUBRICATE	1000 HRS OR 3 YEARS	100 HRS AND/OR NOTE 2
VACUUM RELIEF VALVE FILTER	REPLACE	300 HRS	30 HRS AND/OR NOTE 2
WHEEL BEARING	REPACK AND INSPECT	500 HRS	NOTE 2

Application and Maintenance Schedule Description ⁶

Application et description du calendrier de maintenance⁶

Check one of the following: – Cocher une des réponses suivantes :

- ☐ As a new operator of this aircraft type, the out of phase maintenance intervals specified in STD 625 Appendix C will be used.
À titre de nouvel exploitant de ce type d'aéronef, j'utiliserai les intervalles de maintenance hors calendrier précisés dans l'appendice C de la norme 625
- ☒ As an experienced operator of this aircraft type, or similar types, the out of phase intervals specified in STD 625 Appendix C have been revised as indicated in table 3 or in the documents referenced in table 2
À titre d'exploitant expérimenté de ce type d'aéronef, ou des types semblables, les intervalles hors calendrier précisés dans l'appendice C de la norme 625 ont été révisés comme il est indiqué au tableau 3 ou dans les documents mentionnés au tableau 2

Check one of the following:

Cocher une des réponses suivantes : The maintenance schedules & interim schedules are based upon:

Les calendriers de maintenance et les calendriers provisoires sont fondés sur les textes suivants :

- ☒ STD 625 Appendix B Part 1
Norme 625, appendice B, partie 1
- ☐ The following manufacturer's recommendations
Les calendriers de maintenance et les calendriers provisoires sont fondés sur les textes suivants :

Airframe Document: Document sur la cellule :	Rev #	
	N° de rév	
Engine Document: Document sur les moteurs :	Rev #	
	N° de rév	
Propeller Document: Document sur les hélices :	Rev #	
	N° de rév	
Other Document: Autre document :	Rev #	
	N° de rév	

- ☐ Another operator's maintenance schedule
Calendrier de maintenance d'un autre exploitant

Other operator:
Autre exploitant : _____

Approval no.: _____ (attach program).
N° d'approbation : _____ (joindre le programme)

- ☒ Other data (described below):
Autres données (décrire ci-dessous) :

L'augmentation des TBO moteur est basé sur le Service Instruction 1009BE en date du 2020-04-24.

O-320 D2J et O-360-A4M voir note 11, 15 et 16.

O-320-H2AD voir note 11. (note 15 et 16 ne s'appliquent pas)

AD

⁶ The data on this page is provided for information purposes only, to facilitate Transport Canada evaluation of the schedule.

Les données figurant sur cette page sont fournies à titre d'information seulement pour faciliter l'évaluation du calendrier par Transports Canada.

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2-47. PRE-INSPECTION CHECKS (MODEL 172 AIRPLANES)

A. Pre-inspection Operational Checks.

- (1) Before beginning the step-by-step inspection, start and run up the engine and upon completion, shut down the engine in accordance with instructions in the Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual. During the run-up, observe the following, making note of any discrepancies or abnormalities:
 - (a) Engine temperatures and pressures.
 - (b) Static RPM (Also refer to **Engine** section 11 and 11A of this manual).
 - (c) Magneto drop (Also refer to **Engine** section 11 and 11A of this manual).
 - (d) Engine response to changes in power.
 - (e) Any unusual engine noises.
 - (f) Fuel selector and/or shutoff valve; operate engine on each tank (or cell) position and OFF position long enough to ensure shutoff and/or selector valve functions properly.
 - (g) Idling speed and mixture; proper idle cut-off.
 - (h) Alternator and ammeter.
 - (i) Suction gage.
 - (j) Fuel flow indicator.
- (2) After the inspection has been completed, an engine run-up should again be performed to determine that any discrepancies or abnormalities have been corrected.
- (3) Some of the items in the Inspection Time Limits paragraph are optional, therefore not applicable to all airplanes.

2-48. INSPECTION TIME LIMITS. (MODEL 172 AIRPLANES)

Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES).

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
A Placards (Refer to Pilot's Operating Handbook).						
A	1	Placards and Decals - Inspect presence, legibility, and security. Consult Pilot's Operating Handbook and FAA-Approved Airplane Flight Manual for required placards.		X		
B Fuselage (Section 3).						
B	1	Fuselage Surface - Inspect for skin damage, loose rivets, condition of paint, and check pitot-static ports and drain holes for obstruction. Inspect covers and fairings for security.	X			
B	2	Internal Fuselage Structure – Inspect bulkheads, door posts, stringers, doublers, and skins for corrosion, cracks, buckles, and loose rivets, bolts and nuts.		X		
B	3	Control Wheel Lock - Check general condition and operation.		X		
B	4	Fuselage Mounted Equipment – Check for general condition and security of attachment.		X		
B	5	Antennas and Cables - Inspect for security of attachment, connection, and condition.		X		
B	6	Emergency Locator Transmitter - Inspect for security of attachment and check operation by verifying transmitter output. Check cumulative time and useful life of batteries in accordance with FAR Part 91.207. Refer to Section 16 Emergency Locator Transmitter - Checkout Interval, in the Electrical Systems section of this manual.	X			
B	7	Instrument Panel Shock Mounts, Ground Straps, and Covers – Inspect for deterioration, cracks, and security of attachment.		X		
B	8	Pilot's and Copilot's Inertia Reels - Inspect for security of installation, proper operation, and evidence of damage.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES).

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
B	9	Seats, Seat Belts, and Shoulder Harnesses - Check general condition and security. Check operation of seat stops and adjustment mechanism. Inspect belts for condition and security of fasteners.	X			
B	10	Windows, Windshield, Doors, and Seals - Inspect general condition. Check latches, hinges, and seals for condition, operation, and security of attachment.	X			
B	11	Upholstery, Headliner, Trim, and Carpeting - Check condition and clean as required.			Each 400	Each 1
B	12	Flight Controls - Check freedom of movement and proper operation through full travel with and without flaps extended. Check electric trim controls for operation (as applicable).	X			
B	13	Aileron, Elevator, and Rudder Stops - Check for damage and security.	X			
B	14	Portable Hand Fire Extinguisher - Inspect for proper operating pressure, condition, security of installation, and servicing date.	X			
B	15	Seat Tracks and Stops - Inspect seat tracks for condition and security of installation. Check seat track stops for damage and correct location. Ensure inspection of seat rails for cracks EACH 100 HOURS. Refer to Fuselage Section 3 in this manual.	X			
B	16	Control Column - inspect pulleys, cables, sprockets, bearings, chains, bungees, and turnbuckles for condition.		X		
B	17	Fuel Line and Selector Valve Drain(s) - Remove plug and drain.	X			
B	18	Reel Type Secondary Seat Stops - Make sure manual lock operates correctly.	X			
C	Wings and Empennage (Section 4).					
C	1	Wing Surfaces and Tips – Inspect for skin damage, loose rivets, and condition of paint.	X			
C	2	Wing Struts and Strut Fairings – Check for dents, cracks, loose screws and rivets, and condition of paint.	X			
C	3	Wing Spar and Wing Strut Fittings - Check for evidence of wear Check attach bolts for indications of looseness and retorque as required.		X		
C	4	Wing Structure - Inspect spars, ribs, skins, and stringers for cracks wrinkles loose rivets corrosion or other damage.		X		
C	5	Metal Lines Hoses Clamps and Fittings - Check for leaks condition and security Check for proper routing.		X		
C	6	Wing Access Plates - Check for damage and security of installation.		X		
C	7	Vertical and Horizontal Stabilizers, Tips and Tailcone - Inspect externally for skin damage and condition of paint.	X			
C	8	Vertical and Horizontal Stabilizers and Tailcone Structure - Inspect bulkheads spars, ribs and skins for cracks, wrinkles, loose rivets, corrosion or other damage. Inspect vertical and horizontal stabilizer attach bolts for looseness, retorque as necessary. Check security of inspection covers, fairings and tips.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
D	Landing Gear and Brakes (Section 5).					
D	1	Brakes, Master Cylinders, and Parking Brake - Check master cylinders and parking brake mechanism for condition and security. Check fluid level and test operation of toe and parking brake.	X			
D	2	Main Gear Tubular Struts - Inspect for cracks, dents, corrosion, condition of paint or other damage. Check axles for condition and security.	X			
D	3	Brake Lines Wheel Cylinders, Hoses Clamps and Fittings - Check for leaks, condition, and security and hoses for bulges and deterioration. Check brake lines and hoses for proper routing and support.			Each 400	Each 1
D	4	Wheels, Brake Discs and Linings- Inspect for wear, cracks, warps, dents, or other damage. Check wheel thru-bolts and nuts for looseness.	X			
D	5	Tires- Check tread wear and general condition. Check for proper inflation.	X			
D	6	Wheel Fairings, Strut Fairings, and Cuffs - Check for cracks, dents, and condition of paint.	X		A	
D	7	Main Landing Gear Attachment Structure - Check for damage, cracks, loose rivets, bolts and nuts and security of attachment.	X			
D	8	Nose Gear Steering Mechanism - Check for wear, security, and proper rigging.		X		
D	9	Nose Gear - Inspect torque links, steering rods, and boots for condition and security of attachment. Check strut for evidence of leakage and proper extension. Check strut barrel for corrosion, pitting, and cleanliness. Check shimmy damper and/or bungees for operation, leakage, and attach points for wear and security.	X			
D	10	Nose Gear Fork - Inspect for cracks, general condition, and security of attachment.		X		
D	11	Wheel Bearings - Clean, inspect and lube.			B	
D	12	Nose Gear Attachment Structure - Inspect for cracks, corrosion, or other damage and security of attachment.	X			
E	Aileron Control System (Section 6).					
E	1	Ailerons and Hinges - Check condition, security and operation.	X			
E	2	Aileron Structure, Control Rods, Hinges, Balance Weights, Bellcranks, Linkage, Bolts, Pulleys, and Pulley Brackets - Check condition, operation, and security of attachment.	X			
E	3	Ailerons and Cables - Check operation and security of stops. Check cables for tension, routing, fraying, corrosion, and turnbuckle safety. Check travel if cable tension requires adjustment or if stops are damaged. Check fairleads and rub strips for condition.		X		
E	4	Leave blank				
E	5	Aileron Controls - Check freedom of movement and proper operation through full travel with and without flaps extended.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS	YEARS
F	Wing Flap Control System (Section 7).					
F	1	Flaps - Check tracks, rollers, and control rods for security of attachment. Check operation.	X			
F	2	Flap Actuator Threads - Clean and lubricate. Refer to WING FLAP ACTUATOR lubrication (paragraph 2-43) for detailed instructions.	X			
F	3	Flap Structure, Linkage, Bellcranks, Pulleys, and Pulley Brackets – Check for condition, operation and security.		X		
F	4	Wing Flap Control - Check operation through full travel and observe Flap Position indicator for proper indication.		X		
F	5	Flaps and Cables - Check cables for proper tension, routing, fraying, corrosion, and turnbuckle safety. Check travel if cable tension requires adjustment.		X		
F	6	Flap Motor, Actuator, and Limit Switches (electric flaps) - Check wiring and terminals for condition and security. Check actuator for condition and security.		X		
G	Elevator Control System (Section 8).					
G	1	Elevator Control - Check freedom of movement and proper operation through full travel with and without flaps extended.	X			
G	2	Elevator, Hinges, and Cable Attachment - Check condition, security, and operation.	X			
G	3	Elevator Control System – Inspect pulleys, cables, sprockets, bearings, chains, and turnbuckles for condition, security, and operation.		X		
H	Elevator Trim Tab Control System (Section 9).					
H	1	Elevator Trim Tab and Hinges – Check condition, security, and operation.	X			
H	2	Elevator Trim System - Check cables, push-pull rods, bellcranks, pulleys, turnbuckles, fairleads, rub strips, etc. for proper routing, condition, and security.	X			
H	3	Trim Controls and Indicators – Check freedom of movement and proper operation through full travel. Check pulleys, cables, sprockets, bearings, chains, bungees, and turnbuckles for condition and security. Check electric trim controls for operation as applicable.		X		
H	4	Elevator Trim Tab Stop Blocks – Inspect for damage and security.		X		
H	5	Elevator Trim Tab Actuator – Clean, lubricate, and check free-play.			D	
H	6	Elevator Trim Tab Actuator - Free-Play limits inspection. Refer to Section 9 Elevator Trim Control System section in this manual for cleaning, inspection, and repair procedures.		X		
I	Rudder Control System (Section 10).					
I	1	Rudder - Inspect the rudder skins for cracks and loose rivets, rudder hinges for condition, cracks and security hinge bolts, hinge bearings, hinge attach fittings, and bonding jumper for evidence of damage and wear, failed fasteners, and security. Inspect the rudder hinge bolts for proper safetying of nuts with cotter pins. Inspect balance weight for looseness and the supporting structure for damage.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
I	2	Rudder Pedals and Linkage – Check for general condition, proper rigging, and operation. Check for security of attachment.		X		
I	3	Rudder, Tips, Hinges and Cable Attachment - Check condition, security, and operation.	X			
I	4	Rudder - Check internal surfaces for corrosion, condition of fasteners, and balance weight attachment.		X		
J	Engine (Sections 11 and 11A).					
J	1	Cowling - Inspect for cracks, dents, and other damage, security of cowl fasteners, and cowl mounted landing lights for attachment.	X			
J	2	Engine - Inspect for evidence of oil and fuel leaks. Wash engine and check for security of accessories.	X			
J	3	Engine Controls and Linkage - Check general condition, freedom of movement through full range. Check for proper travel, security of attachment, and for evidence of wear. Check friction locks for proper operation.	X			E
J	4	Ignition Switch and Electrical Harness - Inspect for damage, condition, and security.	X			
J	5	Firewall Structure - Inspect for wrinkles, damage, cracks, sheared rivets, etc. Check cowl shock mounts for condition and security.		X		
J	6	Engine Shock Mounts, Engine Mount Structure, and Ground Straps – Check condition, security, and alignment.		X		
J	7	Induction System - Check security of clamps, tubes, and ducting. Inspect for evidence of leakage.	X			
J	8	Induction Airbox, Valves, Doors, and Controls - Remove air filter and inspect hinges, doors, seals, and attaching parts for wear and security. Check operation. Clean and inspect.	X			
J	9	Induction Air Filter - Remove and clean. Inspect for damage, and service per Paragraph 2-22 and 2-22A. Refer to ENGINE INDUCTION AIR FILTER servicing.	X			F
J	10	Alternate Induction Air System - Check for obstructions, operation, and security.	X			
J	11	Alternator, Mounting Bracket and Electrical Connections – Check condition and security. Check alternator belts for condition and proper adjustment. Check belt tension per Section 16. Paragraph 16-38. Refer to the ALTERNATOR-REMOVAL AND INSTALLATION section in the Electrical Systems section of this manual for the tension check procedure.	X			
J	12	Alternator - Check brushes, leads, commutator or slip ring for wear.				G
J	13	Starter, Starter Solenoid, and Electrical Connections - Check for condition of starter brushes, brush leads, and commutator.	X			H
J	14	Oil Cooler - Check for obstructions, leaks, and security of attachment.	X			
J	15	Exhaust System - Inspect for cracks and security. Special check in area of heat exchanger. Refer to Section 11, Paragraph 11-73, the Engine section in this manual for the inspection procedures.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
J	16	Auxiliary (Electric) Fuel Pump (172Q) - Check pump and fittings for condition, operation, security. Remove and clean filter (as applicable).	X			
J	17	Engine-Driven Fuel Pump – Check for evidence of leakage, security of attachment, and general condition.	X			
J	18	Magnetos - Check external condition, security, and electrical leads for condition. Check timing to engine and internal timing if engine timing requires adjustment.	X		I	
J	19	Magnetos - Check impulse coupling and stop pins for condition, replace as required.			J	
J	20	Magnetos - Inspection, and lubrication procedures.			K	
J	21	Ignition Harness and Insulators – Check for proper routing, deterioration, and condition of terminals.	X			
J	22	Spark Plugs - Remove, clean, analyze, test, gap, and rotate top plugs-to-bottom and bottom plugs-to-top.	X			
J	23	Cylinder Compression – Perform differential compression test.		X		
J	24	Carburetor - Drain and flush carburetor bowl, clean inlet strainer, and drain plug. Check general condition and security.	X			
J	25	Engine Primer - Check for leakage, operation, and security.	X			
J	26	Hoses, Metal Lines, and Fittings - Inspect for signs of oil and fuel leaks. Check for abrasions, chafing, security, proper routing and support and for evidence of deterioration.	X			
J	27	Cold and Hot Air Hoses – Check condition, routing, and security.	X			
J	28	Engine Cylinders, Rocker Box Covers, and Pushrod Housings - Check for fin damage, cracks, oil leakage, security of attachment, and general condition.	X			
J	29	Engine Baffles and Seals – Check condition and security of attachment.	X			
J	30	Crankcase, Oil Sump, and Accessory Section - Inspect for cracks and evidence of oil leakage. Check bolts and nuts for looseness and retorque as necessary. Check crankcase breather lines for obstructions, security, and general condition.	X			
J	31	Engine Oil With Oil Filter - Drain oil sump and oil cooler, replace filter and refill with recommended grade aviation oil.	X		L	
J	32	Engine Oil Without Oil Filter - Drain oil sump and oil cooler, clean and inspect screens, and refill with recommended grade aviation oil.	X		L	
K	Fuel System (Section 12).					
K	1	Fuel Tanks or Integral Fuel Bays - Check for evidence of leakage and condition of fuel caps, adapters, and placards.	X			
K	2	Fuel Tanks or Integral Fuel Bays – Drain fuel and check tank interior and outlet screens.			M	
K	3	Fuel System - Inspect plumbing and components for mounting and security.	X			
K	4	Fuel Tank or Bay Drains - Drain water and sediment.	X			
K	5	Fuel Tank Vent Lines and Vent Valves - Check vents for obstruction and proper positioning.	X			
K	6	Fuel Selector Valve - Check controls for detent in each position, security of attachment, and for proper placarding.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
K	7	Fuel Strainer, Drain Valve, and Controls - Check freedom of movement, security, and proper operation. Disassemble, flush, and clean screen and bowl.	X			
K	8	Fuel Quantity Indicators - Check for damage, security of installation, and perform <u>operational</u> / accuracy test.				Each 1
L	Propeller and Propeller Governor (Section 13).					
L	1	Propeller Mounting - Check for security of installation.	X			
L	2	Propeller Blades - Inspect for cracks, dents, nicks, scratches, erosion, corrosion, or other damage.	X			
L	3	Spinner - Check general condition and attachment.	X			
L	4	Spinner and Spinner Bulkhead - Remove spinner, wash, and inspect for cracks and fractures.	X			
L	5	Propeller Mounting Bolts – Inspect mounting bolts and safety-wire for signs of looseness. Retorque mounting bolts as required.		X		
L	6	Propeller Hub - Check general condition.		X		
M	Utility Systems (Section 14).					
M	1	Ventilation System - Inspect clamps, hoses, and valves for condition and security.			Each 400	Each 1
M	2	Heater Components, Inlets, and Outlets - Inspect all lines, connections, ducts, clamps, seals, and gaskets for condition, restriction, and security.	X			
M	3	Cabin Heat and Ventilation Controls - Check freedom of movement through full travel. Check friction locks for proper operation.		X		
M	4	Pitot Tube and Stall Warning Vane - Check for condition and obstructions.	X			
N	Instruments and Instrument Systems (Section 15).					
N	1	Vacuum System - Inspect for condition and security.	X			
N	2	Vacuum System Hoses - Inspect for hardness, deterioration, looseness, or collapsed hoses.	X			
N	3	Vacuum Pump - Check for condition and security. Check vacuum system breather line for obstructions, condition, and security.	X			
N	4	Vacuum System Air Filter – Inspect for damage, deterioration and contamination. Clean or replace, if required. NOTE: Smoking will cause premature filter clogging.	X		N	
N	5	Vacuum System Relief Valve – Inspect for condition and security.	X		O	
N	6	Instruments - Check general condition and markings for legibility.	X			
N	7	Instrument Lines, Fittings, Ducting, and Instrument Panel Wiring - Check for proper routing, support, and security of attachment.		X		
N	8	Static System - Inspect for security of installation, cleanliness, and evidence of damage.		X		
N	9	Navigation Indicators, Controls, and Components - Inspect for condition and security.		X		
N	10	Airspeed Indicator, Vertical Speed Indicator, - <u>Calibrate</u> Magnetic Compass - Calibrate.				Each 2 Each 1

MODEL 172 SERIES SERVICE MANUAL

Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS YEARS	
N	11	Altimeter and Static System - Inspect in accordance with FAR Part 91.411.				Each 2
N	12	Instrument Panel Mounted Avionics Units (Including Audio Panel, VHF Nav Com(s), ADF, Transponder, DME, and Compass System) - Inspect for deterioration, cracks, and security of instrument panel mounts. Inspect for security of electrical connections, condition, and security of wire routing.		X		
N	13	Avionics Operating Controls – Inspect for security and proper operation of controls and switches and ensure that all digital segments will illuminate properly.		X		
N	14	Remote Mounted Avionics – Inspect for security of units and electrical connectors, condition and security of wire routing. Also, check for evidence of damage and cleanliness.		X		
N	15	Microphones, Headsets, and Jacks - Inspect for cleanliness, security, and evidence of damage.		X		
N	16	Magnetic Compass - Inspect for security of installation, cleanliness, and evidence of damage.		X		
O	Electrical Systems (Section 16).					
O	1	General Airplane and System Wiring - Inspect for proper routing, chafing, broken or loose terminals, general condition, broken or inadequate clamps, and sharp bends in wiring.		X		
O	2	Instrument, Cabin, Navigation, Beacon, Strobe, and Landing Lights – Check operation, condition of lens, and security of attachment.	X			
O	3	Circuit Breakers and Fuses – Check operation and condition. Check for required number of spare fuses.	X			
O	4	Battery - Check general condition and security. Check level of electrolyte.	X		P	
O	5	Battery Box and Cables - Clean and remove any corrosion. Check cables for routing, support, and security of connections.		X		
O	6	Switch and Circuit Breaker Panel, Terminal Blocks, and Junction Boxes - Inspect wiring and terminals for condition and security.		X		
O	7	Alternator Control Unit - Inspect wiring, mounting, condition, and wire routing.		X		
O	8	Switches - Check operation, terminals, wiring, and mounting for conditions, security, and interference.		X		
O	9	Instrument Panel and Control Pedestal - Inspect wiring, mounting, and terminals for condition and security. Check resistance between stationary panel and instrument panel for proper ground.		X		
O	10	External Power Receptacle and Power Cables - Inspect for condition and security.		X		
P	Post Inspection					
P	1	Replace all fairings, doors, and access hole covers. Ground check engine, alternator charging rate, oil pressure, tachometer, oil temperature and pressure gages, and general operation of components.				
Q	Perform the Following Operational Checks:					
Q	1	Brakes - Test toe brakes and parking brake for proper operation.	X			

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Table 1. INSPECTION TIME LIMITS (MODEL 172 AIRPLANES). (Continued)

	S.NO		EACH 100 HOURS	EACH 200 HOURS	SPECIAL INSPECTIONS HOURS	YEARS
R	Service Bulletins/Airworthiness Directives.					
R	1	Check that all applicable Cessna Service Bulletins and Supplier Service Bulletins are complied with.	X	X		
R	2	Check that all applicable Airworthiness Directives and Federal Aviation Regulations are complied with.	X	X		
R	3	Ensure all Maintenance Record Entries required by Federal Aviation Regulations are completed before returning the airplane to service.	X	X		

Special Inspections Legends

- A. If the airplane is flown from surfaces with mud, snow, or ice, the main gear speed fairings should be checked that there is no accumulation which could prevent normal wheel rotation.
- B. Each 500 hours thereafter. More often if operated under prevailing wet or dusty conditions.
- C. Leave blank.
- D. Lubrication of the actuator is required each 1000 hours or 3 years, whichever comes first. See Figure 2-5 for grease specification.
- E. Lubricate each 100 hours (except in extreme dusty conditions). These controls are not repairable and should be replaced every 4800 hours.
- F. Replace 12 Months or 50% surface contamination.
- G. Inspect each 500 hours.
- H. 1. Check solenoid and electrical connections each 100 hours.
2. Inspect the commutator and brushes each 500 hours (not applicable if model Skytec stater install).
- I. If timing to engine is within tolerance - plus zero degrees, minus two degrees, internal timing will not require checking.
- J. Compliance with Lycoming Service Bulletin 425B or latest revision is required. Model 172 with O-320- H2AD engine (1977 thru 1980): Each 500 hours.
- K. 1. Inspect contact points for condition and adjust or replace as required.
2. Inspect carbon brush, high-tension lead, and distributor block for condition and clean or replace as required.
3. Inspect impulse coupling and pawls for condition and replace as required. Use light pressure only, do not force pin (or drill bit) when checking pawls.
4. Inspect bearings and lubricate, replace bearings, if required.
5. Lubricate contact point cam.
6. Leave blank
- L. Model 172 with O-320-H2AD engine (1977 thru 1980): First 25 hours. Drain oil sump, clean pressure screen or replace oil filter, and refill with MIL-L-22851 ashless dispersant oil. If oil must be added during first 25 hours, use MIL-L-6082 aviation grade straight mineral oil. If engine is not equipped with an oil filter, change oil and clean pressure screen at 50 hours, and each 50 hours thereafter. If engine is equipped with an oil filter, change oil and filter at 50 hours, and each 100 hours thereafter. Model 172 with O-320-D2J engine (1981 and ON) and Model 172Q with O-360-A4N engine (1983 and ON): First 25 hours. Drain oil sump, clean pressure screen or replace oil filter, and refill with MIL-L-6080 aviation grade straight mineral oil. If engine is not equipped with an oil filter, change oil and clean pressure screen at 50 hours and each 50 hours there after. If engine is equipped with an oil filter, change oil and filter every 100 hours. Beginning with the 50-hour oil change and thereafter, refill oil sump with MIL-L-22851 ashless dispersant oil.
- M. Each 1000 hours.
- N. Replace every 500 hours.
- O. Replace filter each 300 hours.
- P. Check electrolyte level (if applicable) and clean battery box as required.

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COMPONENT TIME LIMITS (MODEL 172 AIRPLANES).

NOTE: All components not listed herein should be inspected as detailed elsewhere in this chapter and repaired, overhauled or replaced as required. Items shown here should be overhauled or replaced during the regular maintenance periods.

A. Schedule

	COMPONENT	REPLACE (See NOTE 7)	OVERHAUL (See NOTE 7)
A. POWERPLANTS (Sections 11 and 11A).			REFER TO LATEST ISSUE OF MANUFACTURER'S MAINTENANCE MANUAL OR SERVICE BULLETIN
1.	Engines (Avco Lycoming Engine Overhaul Manual) (See NOTE 2)		
2.	Magnetos (Sections 11 and 11A) (See NOTE 3)		
3.	Engine Compartment Flexible Fluid Carrying Rubber Hoses (Cessna Installed) Except Drain Hoses	NOTE 4	
4.	Engine Compartment Drain Hoses	ON CONDITION	
5.	Engine Flexible Hoses (Avco Lycoming Installed) (See Lycoming Engine Maintenance Manual)	NOTE 5	
B. FUEL SYSTEM (Section 12).			
1.	Auxiliary Electric Fuel Pump (See NOTE 6)		
C. PROPELLER (Section 13).			
1.	Propeller (See McCauley Maintenance and Overhaul Manual)		
2.	Governor (See Manufacturer's Manual -McCauley)		
D. NAVIGATION (Section 15).			
1.	Locator Beacon Battery Pack	NOTE 1	
E. ELECTRICAL SYSTEMS (Section 24).			
1.	Landing and Taxi Light Switch or Switch (Landing Light)	NOTE 8	
2.	Switch (Landing Light), Aircraft serials: 17071035 thru 17275034 F17201750 thru F17202134	NOTE 8	
3.	Switch (Beacon Light)	NOTE 9	

NOTE 1: If battery has been in use for more than one collective hour and/or at 50 percent of the useful life of the battery, date on the battery indicates 50 percent of the useful life.

NOTE 2: It is recommended the items listed below be inspected at engine overhaul to establish condition for their replacement or overhaul. Although no overhaul or replacement interval has been established for these items, the inspection of these items at engine overhaul could eliminate overhaul or replacement of the items at a later time.

1. Inspect hoses within the engine compartment in accordance with Engine Compartment Hoses (Refer to Inspection Time Limits). It is recommended the replacement interval not exceed engine overhaul interval.
2. Engine components, such as turbocharger, controller, manifold pressure relief valve and wastegate, magnetos, vacuum pumps, etc., should be inspected for condition, at the time of engine overhaul, as it may be cost effective to overhaul or replace marginal components at that time. A determination is to be made during engine overhaul such that if the components have less hours in service than the engine or have not accumulated hours sufficiently close to the engine overhaul time to warrant overhaul as judged by inspection or the economic aspect, these

MODEL 172 SERIES SERVICE MANUAL

components may not require overhaul or replacement concurrent with engine overhaul. It is recommended that the overhaul or replacement interval for these components not exceed the engine overhaul interval.

3. Inspect the engine compartment for structural damage when engine is removed for overhaul and make the necessary repairs.
4. Inspect the engine exhaust as it may be cost effective to replace marginal components at engine overhaul.
5. Inspect electrical harnesses for damage which would be cost effective to replace at engine overhaul.

NOTE 3: Overhaul Magneto(s) (on condition) at engine overhaul or when engine is partially overhauled for severe environmental affects engine overspeed engine sudden stoppage or other unusual circumstances.

NOTE 4: (This life limit is not intended to allow flexible fluid carrying rubber hoses in a deteriorated or damaged condition to remain in service). Replace engine compartment flexible fluid carrying hoses (Cessna installed only) manufactured of rubber material every 10 years. This does not include drain hoses (on condition).

NOTE 5: Refer to latest Avco Lycoming Engine Service Bulletins.

NOTE 6: Refer to latest Dukes Inc Service Bulletins.

NOTE 7: The terms overhaul and replacement as used within this section dictate action as defined below:

1. Overhaul - Item may be overhauled as defined in Federal Aviation Regulation 43.2 or can be replaced as defined below:
2. Replacement - Item must be replaced with a new item or one that has been rebuilt as defined in Federal Aviation Regulation 43.2.

NOTE 8: On condition

NOTE 9: On condition